EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction, Exothermic And Endothermic Reaction

1. Which represent an exothermic reaction :-

(A) $2HgO\,(s) + 180\ kJ \rightarrow 2Hg(\ell) + O_2(s)$

(B) $N_2O(g) + C(s) \rightarrow CO(g) + N_2(g) - 131\ kJ$

(C) $N_2(g) + O_2\,(s) \rightarrow 2NO(g)$; $\Delta H = +181\ kJ$

(D) $C_2H_2\,(g) + 2H_2(g) \rightarrow C_2H_6(g)$; $\Delta H = -314\ kJ$

CTC086

Relationship Between ΔH And ΔU, ΔH = ΔU + ΔngRT For Chemical Reaction

2. For a reaction, $2X_{(s)} + 2Y_{(s)} \rightarrow 2C_{(l)} + D_{(g)}$

The q_p at 27°C is – 28 Kcal mol^{-1}, the q_V is ..... Kcal mol^{-1}

(A) – 27.4 (B) + 27.4 (C) – 28.6 (D) 28.6

CTC079

3. Consider the reaction at 300 K

$H_{2(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)}$ $\Delta H = -185kJ/mol$

Calculate ΔU if 3 mole of H_2 completely react with 3 mole of Cl_2 to form HCl.

(A) 0 (B) –185 kj (C) 555 kJ (D) – 555 kJ

CTC080

4. The enthalpy of formation of methane(g) at constant pressure is –18,500 cal/mol at 27°C. The enthalpy of formation at constant volume would be

(A) –19,700 cal (B) –17,300 cal (C) –18,498.8 cal (D) –18,500 cal

CTC082

5. For the reaction :

$C_3H_{8\,(g)} + 5O_{2\,(g)} \rightarrow 3CO_{2\,(g)} + 4H_2O_{(l)}$,

value of ΔH – ΔU is :-

(A) +RT (B) –3RT (C) +3RT (D) –RT

CTC170

Enthalpy Of Reaction, Factors Affecting Enthalpy Of Reaction

6. The heat change during the reaction 24 gC and 128 g S following the change $C + 2S \rightarrow CS_2$; $\Delta H = 22$ kcal/mole.

(A) 22 kCal (B) 11 kCal (C) 44 kCal (D) 32 kCal

CTC087

7. Consider the following processes.

$\frac{1}{2}A \longrightarrow B$; $\Delta H\ (kJ/mol^{-1}) = 150$

$3B \longrightarrow 2C + D$; $\Delta H\ (kJ/mol^{-1}) = -125$

$E + A \longrightarrow 2D$; $\Delta H\ (kJ/mol^{-1}) = 350$

For $B + D \longrightarrow E + 2C$; ΔH will be:

(A) 525 kJ mol^{-1} (B) –175 kJ mol^{-1} (C) –325 kJ mol^{-1} (D) 325 kJ mol^{-1}

CTC116

For More Join: @IITJEE_Advanced

Kirchoff's Equation

8. The enthalpy of a reaction at 273 K. is –3.57 KJ. what will be the enthalpy of reaction at 373 K if ΔC_p is zero :–

(A) –3.57 (B) Zero (C) $-3.57 \times \frac{373}{273}$ (D) –375

CTC090

9. The enthalpy changes for the reaction,

$CH_3CHO\ (g) \rightarrow CH_4(g) + CO(g)$ at 300 K is – 17.0 kJ/mol

Calculate the temperature at which $\Delta_r H$ for the reaction will be zero.

[Given : $C_{p,m}\ (CH_4,g) = 38$ J/K mol ; $C_{p,m}\ (CO, g) = 31$ J/K mol & $C_{p,m}\ (CH_3CHO,g) = 52$ J/K mol]

(A) 1300°C (B) 1027°C (C) 700°C (D) 754°C

CTC091

10. Calculate ΔH at 358 K for the reaction

$Fe_2O_3\ (s) + 3H_2\ (g) \longrightarrow 2Fe(s) + 3H_2O(l)$

Given that $\Delta H_{298} = -33.29$ kJ/mol and $C_p\ (Fe_2O_3, s) = 103.8$ J/K mol, $C_p\ (Fe, s) = 25.1$ J/K mol, $C_p\ (H_2O, l) = 75.3$ J/K mol, $C_p\ (H_2, g) = 28.8$ J/K mol.

(A) –33.29 kJ/mol (B) –28.8 kJ/mol (C) –28.136 kJ/mol (D) –33.29 J/mol

CTC092

Enthalpy Of Combustion

11. An exothermic reaction is one in which the reacting substances :

(A) Have same energy as products

(B) Have less energy than the products

(C) Have more energy than the products

(D) Are at higher temperature than the products

Bankers Academy JEE

CTC083

Enthalpy Of Formation

12. Consider the reaction $3O_2 \rightarrow 2O_3$; ΔH = positive,

from the reaction, we can say that :–

(A) Ozone is more stable than oxygen

(B) Ozone is less stable than oxygen and ozone decomposes forming oxygen readily

(C) Oxygen is less stable than ozone and oxygen decomposes forming ozone readily

(D) None of the above

CTC088

13. Given that standard enthalpy of formation of CH_4, C_2H_4 and C_3H_8 are –17.9, 12.5, –24.8 kCal/mol. The ΔH for $CH_4 + C_2H_4 \rightarrow C_3H_8$ at standard conditions is :

(A) – 55.2 kCal (B) – 30.2 kCal (C) 55.2 kCal (D) – 19.4 kCal

CTC093

14. The standard molar heat of formation of ethane, CO_2 and water (ℓ) are respectively –21.1, –94.1 and –68.3 kCal. The standard molar heat of combustion of ethane will be

(A) –372 kCal (B) –162 kCal (C) –240 kCal (D) –183.5 kCal

CTC094

For More Join: @IITJEE_Advanced

15. Two atoms of hydrogen combine to form a molecule of hydrogen gas, the energy of the H_2 molecule is :
(A) Greater than that of separate atoms
(B) Equal to that of separate atoms
(C) Lower than that of separate atoms
(D) Sometimes lower and sometimes higher

CTC095

16. The following are the heats of reactions -

(i) ΔH_f° of $H_2O_{(\ell)} = -68.3$ kCal mol^{-1}

(ii) ΔH_c° of $C_2H_2 = -337.2$ kCal mol^{-1}

(iii) ΔH_c° of $C_2H_4 = -363.7$ kCal mol^{-1}

Then heat change for the reaction $C_2H_2 + H_2 \rightarrow C_2H_4$ is -
(A) –716.1 kCal (B) + 337.2 kCal (C) –41.8 kCal (D) –579.5 kCal

CTC098

Measurement Of ΔU And ΔH, Calorimetry, Bomb Calorimeter

17. The heat capacity of bomb calorimeter is 500 J/°C. A 2°C rise in temperature has been observed on the combustion of 0.1 g of methane. What is the value of ΔE per mole of methane?
(A) 2 kJ (B) 160 kJ (C) –160 kJ (D) –1 kJ

CTC081

Bond Enthalpy

18. The reaction $CH_4(g) + Cl_2(g) \longrightarrow CH_3Cl(g) + HCl(g)$ has $\Delta H = -25$ kcal.

Bond	Bond Enthalpy kCal
C–Cl	84
H–Cl	103
C–H	x
Cl–Cl	y
x : y = 9 : 5	

Rankers Academy JEE

From the given data, what is the bond enthalpy of Cl–Cl bond
(A) 70 kcal/mol (B) 80 kcal/mol (C) 67.75 kcal/mol (D) 57.86 kcal/mol

CTC099

19. For the reaction, $2H_2 + O_2 \longrightarrow 2H_2O$, $\Delta H = -571$ kJ/mol. Bond energy of H – H is 435 kJ/mol, O = O is 498 kJ/mol, then calculate the average bond energy of O – H bond using the above data.
(A) 485 kJ/mol (B) –485 kJ/mol (C) 271 kJ/mol (D) –271 kJ/mol

CTC114

Ionisation Enthalpy

20. The ionization energy of an element is
(A) the same as the electron affinity of the element
(B) equal in magnitude but of opposite sign to the electron affinity of the element
(C) the energy released when an electron is added to an atom of the element
(D) the energy required to remove the outermost electron of an atom of the element

CTC101

Lattice Enthalpy

21. Ionic compounds are formed most easily with
(A) low electron affinity, high ionisation energy
(B) high electron affinity, low ionisation energy
(C) low electron affinity, low ionisation energy
(D) high electron affinity, high ionisation energy

CTC104

For More Join: @IITJEE_Advanced

Born Haber Cycle

22. Calculate the lattice energy of NaI crystal from the following data by the use of Born-Haber Cycle.
Heat of formation of NaI = –287.6 kJ/mol

$Na(s) \rightarrow Na(g)$; $\Delta H = 108.7$ kJ/mol

$Na(g) \rightarrow Na^{\oplus}(g) + e^{\ominus}$; $\Delta H = 493.8$ kJ/mol

$I_2 \rightarrow I(g)$; $\frac{1}{2}\Delta H_D = 106.6$ kJ/mol

$I(g) + e^{\ominus} \rightarrow I^{\ominus}(g)$; $\Delta H = -305.9$ kJ/mol

(A) –603.4 kJ/mol (B) –690.8 kJ/mol
(C) 609.4 kJ/mol (D) 600 kJ/mol

CTC106

Enthalpy Of Solution, Enthalpy Of Dilution

23. The enthalpy of solution of sodium chloride is 4 kJ/mol and its enthalpy of hydration of ions is –784 kJ/mol. Then the lattice enthalpy of NaCl (in kJ/mol) is

(A) + 788 (B) + 4 (C) + 398 (D) +780

CTC107

24. The dissolution of $CaCl_2.6H_2O$ in a large volume of water is endothermic to the extent of 3.5 kcal/mol. For the reaction $CaCl_2(s) + 6H_2O(\ell) \rightarrow CaCl_2.6H_2O(s)$; ΔH is –23.2 kcal. The heat of solution of anhydrous $CaCl_2$ in large quantity of water will be

(A) –26.7 kcal mol^{-1} (B) –19.7 kcal mol^{-1}
(C) +19.7 kcal mol^{-1} (D) +26.7 kcal mol^{-1}

CTC108

Enthalpy Of Neutralisation

25. Which of the following reaction is endothermic

(A) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$
(B) $Fe(s) + S(s) \longrightarrow FeS(s)$
(C) $NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(\ell)$
(D) $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell)$

Narayana Academy JEE

CTC085

26. If $H^{\oplus} + OH^{\ominus} \rightarrow H_2O + 13.7$ kcal, then heat of complete neutralisation of one gram mole of H_2SO_4 with strong base will be :

(A) 13.7 kcal (B) 27.4 kcal (C) 6.85 kcal (D) 3.425 kcal

CTC109

27. What would be the heat released when an aqueous solution containing 0.5 mole of HNO_3 is mixed with 0.3 mol of $OH^{\ominus}$ (enthalpy of neutralisation = – 57.1 kJ/mol)?

(A) 28.5 kJ (B) 17.13 kJ (C) 45.7 kJ (D) 1.7 kJ

CTC115

Resonance Energy

28. If bond enthalpy of C–C and C = C are 348 kJ/mole and 615 kJ/mole respectively then calculate enthalpy change (in kJ/mole) which occurs during the isomerisation of cyclopropane (g) into propene(g)

(A) –267 (B) 81 (C) –81 (D) 267

CTC100

29. For a given reaction, $C_7H_8(l) + 9O_2(g) \longrightarrow 7CO_2(g) + 4H_2O(l)$, the calculated and observed heats of reaction are respectively 232 and 50.4 kJ mol^{-1}. The resonance energy for it will be :

(A) 160 kJ mol^{-1} (B) +172.2 kJ mol^{-1}
(C) +182.2 kJ mol^{-1} (D) –182.2 kJ mol^{-1}

CTC117

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Enthalpy Of Formation

1. From the following data at 25°C, which of the following statement(s) is/are correct?

$\frac{1}{2}H_2(g) + \frac{1}{2}O_2(g) \longrightarrow OH(g)$: $\Delta H^o = 42$ kJ

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g)$: $\Delta H^o = -242$ kJ

$H_2(g) \longrightarrow 2H(g)$: $\Delta H^o = 436$ kJ

$O_2(g) \longrightarrow 2O(g)$: $\Delta H^o = 495$ kJ

(A) $\Delta_r H°$ for the reaction $H_2O(g) \longrightarrow 2H(g) + O(g)$ is 925.5 kJ.

(B) $\Delta_r H°$ for the reaction $OH(g) \longrightarrow H(g) + O(g)$ is 502 kJ.

(C) Enthalpy of formation of H(g) is −218 kJ/mol.

(D) Enthalpy of formation of OH(g) is 42kJ/mol.

CTC150

Rankers Academy JEE

Electron Gain Enthalpy

2. The enthalpy change for the following process would be expected to be a negative for

(A) $Na^+(g) + e^- \rightarrow Na(g)$ (B) $F(g) + e^- \rightarrow F^-(g)$

(C) $Na^+(g) + F^-(g) \rightarrow NaF(s)$ (D) $H_2O(l) \rightarrow H_2O(g)$

CTC151

3. α-maltose can be hydrolysed to glucose according to the following reaction.

$\alpha\text{-}C_{12}H_{22}O_{11}(aq) + H_2O(l) \longrightarrow 2C_6H_{12}O_6(aq)$

The standard enthalpy of formation of $H_2O(l)$, $C_6H_{12}O_6(aq)$ and $\alpha\text{–}C_{12}H_{22}O_{11}(aq)$ are −285, −1263 and −2238 kJ/mol, respectively. Which of the following statement(s) is/are true?

(A) The hydrolysis reaction is exothermic.

(B) Heat liberated in combustion of 1.0mole of α–maltose is smaller than the heat liberated in combustion of 2.0 mole of glucose neglecting the effect of hydration.

(C) Increase in temperature will increase the degree of hydrolysis of α-maltose.

(D) Enthalpy of reaction will remain the same even if solid α-maltose is taken in the reaction.

CTC152

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Born Haber Cycle

Comprehension # (Q.4 to Q.6)

Use the data (all values are in kJ per mole at 25°C) given below to answer the following.
Enthalpy of formation of CH_3CN = +88.0
Enthalpy of formation of C_3H_8 = −85.0
Enthalpy of sublimation of graphite = 719.0
Enthalpy of dissociation of nitrogen = 948.0
Enthalpy of dissociation of hydrogen = 435.0
Bond enthalpies: C–H = 414.0; C–N = 378.0; N–H = 426.0

4. The bond enthalpy of C–C bond (kJ/mol) is
(A) 250.0 (B) 335.0 (C) 223.33 (D) 248.5

CTC153

5. The bond enthalpy of C≡N bond (kJ/mol)_is
(A) 987.5 (B) 811.5 (C) 899.5 (D) 890.0

CTC154

6. The enthalpy of hydrogenation of CH_3CN (kJ/mol) is
(A) −288.5 (B) +288.5 (C) −89.5 (D) +89.5

CTC155

Pankeers Academy JEE

Enthalpy Of Solution, Enthalpy Of Dilution

Comprehension # (Q.7 to 8)

The thermal effects of reactions in liquid NH_3 at −33°C were measured by observing the quantity of liquid NH_3 vaporized by the process of interest. The heat of vaporization of NH_3 at −33°C is 320.0 cal/g. When 0.98 g of NH_4Br was dissolved in 20 g of liquid NH_3, 0.25 g of NH_3 was vaporized (Br = 80).

7. The molar heat of solution of NH_4Br in liquid NH_3 at this concentration is
(A) +80.0 cal (B) – 80.0 cal (C) – 8.0 kcal (D) +8.0 kcal

CTC156

8. When 0.49 g of NH_4Br was dissolved in 20 g of liquid NH_3 containing an equimolar amount of KNH_2, 0.80 g of ammonia was vaporized. The ΔH for the reaction

$NH_4^+(NH_3, l) + NH_2^-(NH_3, l) \rightarrow 2NH_3(l)$ at 240 K is
(A) −256.0 cal (B) +256.0 cal (C) +51.2 kcal (D) −51.2 kcal

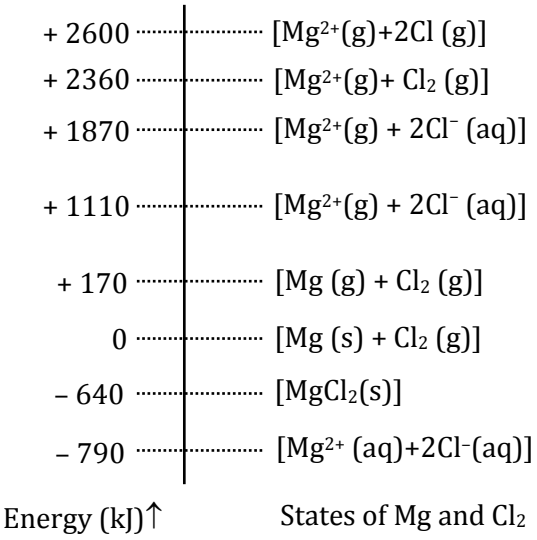
CTC157

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

Born Haber Cycle

9. Carefully observe the given diagrams which indicate the standard enthalpy of formation of different states of one mole of Mg and two moles of Cl atoms and match the entries in Column I and II provided.



Column -I		Column -II	
(A)	$\Delta_f H$ [Mg^{2+} (aq)]	(P)	–1900 kJ/mol
(B)	$\Delta_f H$ [Cl^- (aq)]	(Q)	+460 kJ/mol
(C)	$\Delta H_{Hydration}$ [Mg^{2+}(g)]	(R)	–625 kJ/mol
(D)	Lattice enthalpy of $MgCl_2$(s)	(S)	+2510 kJ/mol

CTC158

Enthalpy Of Neutralisation

10. Column I contains some chemical reactions and Column II contains some ΔH values (in kJ). Match the correct ΔH in Column II for the chemical reaction in Column I with the help of the following thermochemical equations given.

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$; $\Delta H = -57.3$ kJ

$\Delta H_{Solution}$ of HA(g) = –70.7 kJ/mol

$\Delta H_{Solution}$ of BOH(g) = 20 kJ/mol

$\Delta H_{Ionization}$ of HA = 15 kJ/mol and BOH is a strong base.

Column -I (Chemical reactions)		Column -II (ΔH values in kJ)	
(A)	$HA(aq) + BOH(aq) \rightarrow BA(aq) + H_2O$	(P)	–42.3
(B)	$HA(g) + BOH(g) \rightarrow BA(aq) + H_2O$	(Q)	–93
(C)	$HA(g) \rightarrow H^+(aq) + A^-(aq)$	(R)	–55.7
(D)	$B^+(aq) + OH^-(aq) \rightarrow OH(aq)$	(S)	0

CTC159

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Enthalpy Of Formation

1. Find standard enthalpy of formation of ammonia gas in kJ/mol -

Given : $N_2H_4(g) + H_2(g) \longrightarrow 2NH_3(g)$; $\Delta_rH^\circ = -40 kJ/mol$

$\Delta H_f^\circ[N_2H_4(g)] = -120 kJ/mol$

CTC023

2. The standard enthalpy of neutralization of KOH with (a) HCN (b) HCl in dilute solution is –2480 cal/mol and –13.68 kcal/mol respectively. Find the enthalpy of dissociation of HCN at the same temperature in kJ mol^{-1}.

CTC028

3. At 300K, the standard enthalpies of formation of $C_6H_5COOH(s)$, $CO_2(g)$ & $H_2O(l)$ are; –408, –393 & –286 kJ mol^{-1} respectively. Calculate the heat of combustion of benzoic acid at constant volume (in kJ mol^{-1}).

CTC029

Rankers Academy JEE

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Enthalpy Of Formation

4. The enthalpy change for the reaction $C_3H_8(g) + H_2(g) \longrightarrow C_2H_6(g) + CH_4(g)$ at 25° C is – 55.7 kJ/mol. Calculate the enthalpy of combustion of $C_2H_6(g)$ (in kJ/mole). The enthalpy of combustion of H_2, C_3H_8 & CH_4 are –285.7, – 2220 & –890.0 kJ/mol respectively.

CTC119

5. Calculate standard enthalpy of formation of carbon–di–sulphide (*l*) (in kJ/mole). Given the standard enthalpy of combustion of carbon (s) , sulphur (s) & carbon–di–sulphide (*l*) are : – 393, – 293 and – 1108 kJ mol^{-1} respectively.

CTC120

6. When 2 moles of $C_2H_6(g)$ are completely burnt, 3120 kJ of heat is liberated. Calculate the enthalpy of formation (in kJ/mol) of $C_2H_6(g)$. Given Δ_fH for $CO_2(g)$ & H_2O (*l*) are – 395 & – 286 kJ respectively.

CTC121

For More Join: @IITJEE_Advanced

Bond Enthalpy

7. The polymerisation of ethylene to linear polyethylene is represented by the reaction

$nCH_2 = CH_2(g) \rightarrow (-CH_2-CH_2)_n(g)$

where n has a large integral value . Given that the average enthalpies of bond dissociation for C=C & C–C at 298 K are + 590 & + 331 kJ mol^{-1} respectively. Calculate the enthalpy of polymerisation per mole of ethylene (in kJ) at 298 K .

CTC123

Enthalpy Change During Phase Transformations (Fusion, Vapourisation, Sublimation)

8. Compute the enthalpy of formation of liquid methyl alcohol in kJ mol^{-1}, using the following data. Enthalpy of vaporisation of liquid CH_3OH = 38 kJ/ mol .

Enthalpy of formation of gaseous atoms from the elements in their standard states are

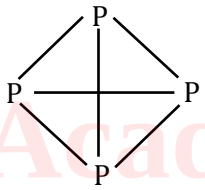
$H \rightarrow 218$ kJ / mol ; $C \rightarrow 715$ kJ / mol ; $O \rightarrow 249$ kJ / mol.

Bond Enthalpies

C – H → 415 kJ / mol ; C – O → 356 kJ / mol ; O – H → 463 kJ / mol

CTC122

9. White phosphorus is a tetra-atomic solid $P_4(s)$ at room temperature.



Rankers Academy JEE

Find average (P –P) bond enthalpy in kJ/mol.

Given : $\Delta H_{sublimation}$ of P_4 (s) = 59 kJ/mol

$\Delta H_{atomisation}$ of P_4 (s) = 1265 kJ/mol

CTC124

Born Haber Cycle

10. Calculate the electron gain enthalpy of fluorine atom (in kJ/mole) using the following data. (All the values are in kJ mol^{-1} at 25° C). ΔH_{diss} (F_2) = 160, $\Delta_f H$ (NaF(s)) = – 571, I.E. [Na(g)] = 494, ΔH_{sub} [Na(s)] = 101 , Lattice enthalpy of NaF(s) = 894.

CTC127

Enthalpy Of Solution, Enthalpy Of Dilution

11. The enthalpy of solution of anhydrous $CuSO_4$ is –16 kcal/mol and that of $CuSO_4.5H_2O$ is 3 kcal/mol Calculate the enthalpy of hydration of $CuSO_4$ (in kcal/mol)

CTC128

12. If the enthalpy of formation of HCl (g) and Cl^- (aq) are –90 kJ/mole and –170 kJ/mol, find the enthalpy of solution of hydrogen chloride gas (in kJ/mol).

CTC129

For More Join: @IITJEE_Advanced

Enthalpy Of Neutralisation

13. Enthalpy of neutralization of H_3PO_3 acid is -106.68 kJ/mol using NaOH. If enthalpy of neutralization of HCl by NaOH is –55.84 kJ/mol. Calculate $\Delta H_{ionization}$ of H_3PO_3 into its ions (in kJ/mol)

CTC130

Resonance Energy

14. Calculate enthalpy of combustion of propane [C_3H_8 (g)] in kJ/mol at 298 K.

Given : B.E. (O = O) = 498 kJ/mole ; B.E. (C = O) = 804 kJ/mole

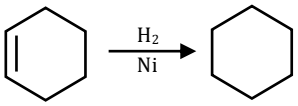
B.E. (C–H) = 410 kJ/mole ; B.E. (O–H) = 464 kJ/mole

B.E. (C–C) = 345 kJ/mole ; Resonance energy of CO_2 (g) = – 143 kJ/mole

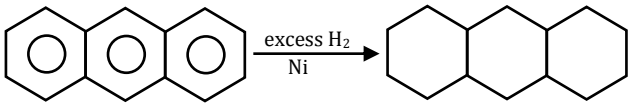
$\Delta H_{vaporization}$ (H_2O, *l*) = 41 kJ/mole

CTC125

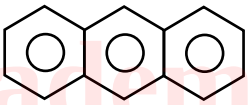
15. Use the following data to answer the questions below :



; $\Delta H = -28.6$ Kcal mol^{-1} ;



; $\Delta H = -116.2$ Kcal mol^{-1}

Calculate the resonance energy of anthracene, 

Rankers Academy JEE

CTC126

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. For the reaction
$H_2F_2(g) \rightarrow H_2(g) + F_2(g)$; $\Delta U = -59.6$ kJ mol^{-1} at 27°C.
The enthalpy change for the above reaction is (–) ___ kJ mol^{-1} [nearest integer]
Given : R = 8.314 JK^{-1} mol^{-1}. **[JEE (Main) 2022]**
CTC171

2. Solid fuel used in rocket is a mixture of Fe_2O_3 and Al (in ratio 1 : 2). The heat evolved (kJ) per gram of the mixture is _____ (Neatest integer)
Given : $\Delta H_f^{\theta}(Al_2O_3) = -1700\,kJ\,mol^{-1}$; $\Delta H_f^{\theta}(Fe_2O_3) = -840\,kJ\,mol^{-1}$
Molar mass of Fe, Al and O are 56, 27 and 16 g mol^{-1} respectively.
[JEE (Main) 2023]
CTC133

3. An athlete is given 100 g of glucose ($C_6H_{12}O_6$) for energy. This is equivalent to 1800 kJ of energy. The 50% of this energy gained is utilized by the athlete for sports activities at the event. In order to avoid storage of energy, the weight of extra water he would need to perspire is _____ g (Nearest integer)
Assume that there is no other way of consuming stored energy.
Given : The enthalpy of evaporation of water is 45 kJ mol^{-1}
Molar mass of C, H & O are 12. 1 and 16 g mol^{-1}. **[JEE (Main) 2023]**
CTC160

4. Consider the following cases of standard enthalpy of reaction $\left(\Delta H_r^{\circ}\,in\,kJ\,mol^{-1}\right)$

Pankers Academy JEE

$$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(\ell)\ \Delta H_1^{\circ} = -1550$$

$$C(graphite) + O_2(g) \rightarrow CO_2(g)\ \Delta H_2^{\circ} = -393.5$$

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)\ \Delta H_3^{\circ} = -286$$

The magnitude of $\Delta H_{f\,C_2H_6(g)}^{\circ}$ is _____ kJ mol^{-1} (Nearest integer). **[JEE (Main) 2025]**
CTC189

5. $S(g) + \frac{3}{2}O_2(g) \rightarrow SO_3(g) + 2x$ kcal

$SO_2(g) + \frac{1}{2}O_2(g) \rightarrow SO_3(g) + y$ kcal

The heat of formation of $SO_2(g)$ is given by : **[JEE (Main) 2025]**

(1) $\frac{2x}{y}$ kcal (2) y – 2x kcal (3) 2x + y kcal (4) x + y kcal

CTC190

6. Consider the following data :
Heat of formation of $CO_2(g)$ = –393.5 kJ mol^{-1}
Heat of formation of $H_2O(l)$ = – 286.0 kJ mol^{-1}
Heat of combustion of benzene = –3267.0 kJ mol^{-1}
The heat of formation of benzene is ____ kJ mol^{-1}.
(Nearest integer) **[JEE (Main) 2025]**
CTC191

For More Join: @IITJEE_Advanced

7. Combustion of 1 mole of benzene is expressed at

$C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow CO_2(g) + 3H_2O(l)$.

The standard enthalpy of combustion of 2 mol of benzene is – 'x' kJ.

x = _____.

(1) standard Enthalpy of formation of 1 mol of $C_6H_6(l)$, for the reaction $6C(graphite) + 3H_2(g) \rightarrow C_6H_6(l)$ is 48.5 kJ mol^{-1}.

(2) Standard Enthalpy of formation of 1 mol of $CO_2(g)$, for the reaction $C(graphite) + O_{2(g)} \rightarrow CO_2(g)$ is –393.5 kJ mol^{-1}.

(3) Standard and Enthalpy of formation of 1 mol of $H_2O(l)$, for the reaction $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ is –286 kJ mol^{-1}.

[JEE (Main) 2024]

CTC161

8. 0.3 g of ethane undergoes combustion at 27°C in a bomb calorimeter. The temperature of calorimeter system (including the water) is found to rise by 0.5°C. The heat evolved during combustion of ethane at constant pressure is _____kJ mol^{-1}. (Nearest integer)

[Given : The heat capacity of the calorimeter system is 20 kJ K^{-1}, R = 8.3 JK^{-1} mol^{-1}.

Assume ideal gas behaviour.

Atomic mass of C and H are 12 and 1 g mol^{-1} respectively] **[JEE (Main) 2023]**

CTC163

9. The reaction of cyanamide, $NH_2CN_{(s)}$ with oxygen was run in a bomb calorimeter and ΔU was found to be –742.24 kJ mol^{1-}. The magnitude of ΔH_{298} for the reaction

$NH_2CN(s) + \frac{3}{2}O_2(g) \rightarrow N_2(g) + CO_2(g) + H_2O(\ell)$ is ___ kJ. (Rounded off to the nearest integer)

[Assume ideal gases and R = 8.314 J mol^{-1} K^{-1}] **[JEE (Main) 2021]**

CTC172

10. The bond dissociation enthalpy of X_2 ΔH°_{bond} calculated from the given data is _____ kJ mol^{-1}. (Nearest integer)

$M^+X^-(s) \rightarrow M^+(g) + X^-(g)$ $\Delta H^{\circ}_{lattice} = 800$ kJ mol^{-1}

$M(s) \rightarrow M(g)$ $\Delta H^{\circ}_{sub} = 100$ kJ mol^{-1}

$M(g) \rightarrow M^+(g)^- + e^-(g)$ $\Delta H^{\circ}_{i} = 500$ kJ mol^{-1}

$X(g) + e^-(g) \rightarrow X^-(g)$ $\Delta H^{\circ}_{eg} = -300$ kJ mol^{-1}

$M(s) + \frac{1}{2}X_2(g) \rightarrow M^+X^-(s)$ $\Delta H^{\circ}_{f} = -400$ kJ mol^{-1}

[Given : M^+X^- is a pure ionic compound and X forms a diatomic molecule X_2 is gaseous state]

[JEE (Main) 2025]

CTC192

11. The formation enthalpies, $\Delta H^{\ominus}_f$ for $H_{(g)}$ and $O_{(g)}$ are 220.0 and 250.0 kJ mol^{-1}, respectively, at 298.15 K, and $\Delta H^{\ominus}_f$ for $H_2O_{(g)}$ is –242.0 kJ mol^{-1} at the same temperature. The average bond enthalpy of the O–H bond in water at 298.15 K is _____ kJ mol^{-1} (nearest integer). **[JEE (Main) 2025]**

CTC193

12. The enthalpy of formation of ethane (C_2H_6) from ethylene by addition of hydrogen where the bond-energies of C – H, C – C, H – H are 414 kJ, 347 kJ, 615 kJ and 435 kJ respectively is - _____ kJ.

[JEE (Main) 2024]

CTC164

For More Join: @IITJEE_Advanced

13. $A_2 + B_2 \rightarrow 2AB; \Delta H_f^0 = -200\ kJmol^{-1}$

AB, A_2 and B_2 are diatomic molecule. If the bond enthalpies of A_2, B_2 and AB are in the ratio 1:0.5:1, then the bond enthalpy of A_2 is _____ kJ mol^{-1} (Nearest integer)

[JEE (Main) 2023]

CTC165

14. If enthalpy of atomisation for $Br_{2(\ell)}$ is x kJ/mol and bond enthalpy for Br_2 is y kJ/mol, the relation between them : **[JEE (Main) 2020]**

(1) is x = y (2) is x < y (3) does not exist (4) is x > y

CTC044

15. Standard enthalpy of vapourisation for CCl_4 is 30.5 kJ mol^{-1}. Heat required for vapourisation of 284 g of CCl_4 at constant temperature is ___kJ.

(Given molar mass in g mol^{-1} ; C = 12, Cl = 35.5) **[JEE (Main) 2024]**

CTC166

16. Enthalpy of sublimation of iodine is 24 cal g^{-1} at 200°C. If specific heat of $I_2(s)$ and $I_2(vap)$ are 0.055 and 0.031 cal $g^{-1}K^{-1}$ respectively, then enthalpy of sublimation of iodine at 250°C in cal g^{-1} is:

[JEE (Main) 2019]

(1) 2.85 (2) 11.4 (3) 5.7 (4) 22.8

CTC042

17. If C(diamond)→C(graphite) + X kJ mol^{-1}

C(diamond) + $O_2(g) \rightarrow CO_2(g)$ + Y kJ mol^{-1}

C(graphite) + $O_2(g) \rightarrow CO_2(g)$ + Z kJ mol^{-1}

At constant temperature. Then **[JEE (Main) 2025]**

(1) X = Y + Z (2) –X = Y + Z (3) X = –Y + Z (4) X = Y – Z

CTC194

18. The Born-Haber cycle for KCl is evaluated with the following data :

$\Delta_f H^\circ$ for KCl = $-436.7\ kJ\ mol^{-1}$;

$\Delta_{sub} H^\circ$ for K = 89.2 kJ mol^{-1};

$\Delta_{ionization} H^\circ$ for K=419.0 kJ mol^{-1} ; $\Delta_{EGE} H^\circ$ for $Cl_{(g)}$ = –348.6 kJ mol^{-1} ; Δ_{bond} for Cl_2 = 243.0 kJ mol^{-1}

The magnitude of lattice enthalpy of KCl in kJ mol^{-1} is _____ (Nearest integer) **[JEE (Main) 2021]**

CTC047

19. The heat of solution of anhydrous $CuSO_4$ and $CuSO_4 \cdot 5H_2O$ are –70 kJ mol^{-1} and +12 kJ mol^{-1} respectively.

The heat of hydration of $CuSO_4$ to $CuSO_4 \cdot 5H_2O$ is –x kJ. The value of x is____. **[JEE (Main) 2024]**

CTC167

20. The number of endothermic process/es from the following is _____

A. $I_2(g) \rightarrow 2I(g)$

B. $HCl(g) \rightarrow H(g) + Cl(g)$

C. $H_2O(l) \rightarrow H_2O(g)$

D. $C(s) + O_2(g) \rightarrow CO_2(g)$

E. Dissolution of ammonium chloride in water **[JEE (Main) 2023]**

CTC132

Rankers Academy JEE
For More Join: @IITJEE_Advanced

21. When equal volume of 1M HCl and 1M H_2SO_4 are separately neutralised by excess volume of 1M NaOH solution. X and y kJ of heat is liberated respectively. The value of y/x is _____.

[JEE (Main) 2024]

CTC168

22. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

Assertion (A): Enthalpy of neutralisation of strong monobasic acid with strong monoacidic base is always –57 kJ mol^{-1}

Reason (R): Enthalpy of neutralisation is the amount of heat liberated when one mole of H^+ ions furnished by acid combine with one mole of ^-OH ions furnished by base to form one mole of water.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) **(A)** is true but **(R)** is false

(2) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(3) **(A)** is false but **(R)** is true

(4) Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**

[JEE (Main) 2024]

CTC169

23. While performing a thermodynamics experiment, a student made the following observations,

$HCl + NaOH \rightarrow NaCl + H_2O$ $\Delta H = -57.3$ kJ mol^{-1}

$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

$\Delta H = -55.3$ kJ mol^{-1}.

The enthalpy of ionization of CH_3COOH as calculated by the student is _____ kJ mol^{-1}. (nearest integer)

[JEE (Main) 2022]

CTC048

Rankers Academy JEE

24. 200mL of 0.2M HCl is mixed with 300 mL of 0.1 M NaOH. The molar heat of neutralization of this reaction is –57.1 kJ. The increase in temperature in °C of the system on mixing is $x \times 10^{-2}$.

The value of x is _____. (Nearest integer)

[Given : Specific heat water = 4.18 J $g^{-1}K^{-1}$

Density of water = 1.00 g cm^{-3}]

(Assume no volume change on mixing)

[JEE (Main) 2021]

CTC173

25. Which of the following mixing of 1M base and 1M acid leads to the largest increase in temperature?

[JEE (Main) 2025]

(1) 30 mL HCl and 30 mL NaOH

(2) 30 mL CH_3COOH and 30 mL NaOH

(3) 50 mL HCl and 20 mL NaOH

(4) 45 mL CH_3COOH and 25 mL NaOH

CTC195

For More Join: @IITJEE_Advanced

EXERCISE – JEE (Advanced) PYQ

Paragraph For question # 01 and 02

The amount of energy required to break a bond is same as the amount of energy released when the same bond is formed. In gaseous state, the energy required for *homolytic cleavage* of a bond is called Bond Dissociation Energy (BDE) or Bond Strength. BDE is affected by *s*-character of the bond and the stability of the radicals formed. Shorter bonds are typically stronger bonds. BDEs for some bonds are given below:

$H_3C—H_{(g)} \longrightarrow H_3\dot{C}_{(g)} + \dot{H}(g)$ $\Delta H°=105$kcal mol^{-1}

$Cl—Cl_{(g)} \longrightarrow \dot{Cl}_{(g)} + \dot{Cl}(g)$ $\Delta H°=58$kcal mol^{-1}

$H_3C—Cl_{(g)} \longrightarrow H_3\dot{C}_{(g)} + \dot{Cl}(g)$ $\Delta H°=85$kcal mol^{-1}

$H—Cl_{(g)} \longrightarrow \dot{H}_{(g)} + \dot{Cl}(g)$ $\Delta H°=103$kcal mol^{-1}

1. Correct match of the C–H bonds (shown in bold) in Column J with their BDE in Column K is

Column- J Molecule	**Column- K** BDE (kcal mol^{-1})
(P) H–$CH(CH_3)_2$	(i) 132
(Q) H–CH_2Ph	(ii) 110
(R) H–$CH=CH_2$	(iii) 95
(S) H–C≡CH	(iv) 88

Rankers' Academy JEE

[JEE (Advanced) 2021]

(A) P – iii, Q – iv, R – ii, S – i
(B) P – i, Q – ii, R – iii, S – iv
(C) P – iii, Q – ii, R –i, S – iv
(D) P – ii, Q – i, R – iv, S – iii

CTC076

2. For the following reaction

$CH_4(g) + Cl_2(g) \xrightarrow{\text{light}} CH_3Cl(g) + HCl\ (g)$

the correct statement is **[JEE (Advanced) 2021]**

(A) Initiation step is exothermic with $\Delta H° = -58$ kcal mol^{-1}

(B) Propagation step involving $^{\bullet}CH_3$ formation is exothermic with $\Delta H° = -2$ kcal mol^{-1}.

(C) Propagation step involving CH_3Cl formation is endothermic with $\Delta H° = +27$ kcal mol^{-1}.

(D) The reaction is exothermic with $\Delta H° = -25$ kcal mol^{-1}.

CTC077

3. Using the data provided, calculate the multiple bond energy (kJ mol^{-1}) of a C ≡ C bond in C_2H_2. (Take the bond energy of a C–H bond as 350 kJ mol^{-1}.)

$2\ C(s) + H_2(g) \longrightarrow C_2H_2(g); \Delta H = 225$ kJ mol^{-1}

$2\ C(s) \longrightarrow 2C(g);\ \Delta H = 1410$ kJ mol^{-1}

$H_2(g) \longrightarrow 2H(g);\ \Delta H = 330$ kJ mol^{-1} **[JEE (Advanced) 2012]**

(A) 1165 (B) 837 (C) 865 (D) 815

CTC071

For More Join: @IITJEE_Advanced

4. The standard enthalpies of formation of $CO_2(g)$, $H_2O(l)$ and glucose(s) at 25°C are –400 kJ/mol, –300 kJ/mol & –1300 kJ/mol, respectively. The standard enthalpy of combustion per gram of glucose at 25°C is –

[JEE (Advanced) 2013]

(A) + 2900 kJ (B) – 2900 kJ (C) –16.11 kJ (D) + 16.11 kJ

CTC072

5. Choose the reaction(s) from the following options, for which the standard enthalpy of reaction is equal to the standard enthalpy of formation of the product.

[JEE (Advanced) 2019]

(A) $\frac{3}{2}O_2(g) \rightarrow O_3(g)$ (B) $\frac{1}{8}S_8(s) + O_2(g) \rightarrow SO_2(g)$

(C) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$ (D) $2C(g) + 3H_2(g) \rightarrow C_2H_6(g)$

CTC075

6. 2 mol of Hg(g) is combusted in a fixed volume bomb calorimeter with excess of O_2 at 298 K and 1 atm into HgO(s). During the reaction, temperature increases from 298.0 K to 312.8 K. If heat capacity of the bomb calorimeter and enthalpy of formation of Hg(g) are 20.00 kJ K^{-1} and 61.32 kJ mol^{-1} at 298 K, respectively, the calculated standard molar enthalpy of formation of HgO(s) at 298 K is X kJ mol^{-1}. The value of |X| is _____.

[**Given :** Gas constant R = 8.3 J K^{-1} mol^{-1}]

[JEE (Advanced) 2022]

CTC078

Rankers Academy JEE

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	D	B	B	C	B	A	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	D	A	C	C	C	D	A	D
Que.	21	22	23	24	25	26	27	28	29	
Ans.	B	B	A	B	A	B	B	B	D	

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	
Ans.	AD	ABC	AB	B	C	A	C	D	
Que.	9				10				
Ans.	A → Q; B → R; C → P; D → S				A → P, B → Q; C → R; D → S				

EXERCISE (S)

1. –80 **2.** 11 **3.** –3201 **4.** –1560.00 **5.** 129.00

6. –88.00 **7.** –72 **8.** –266 **9.** 201.00 **10.** –352.00

11. –19.00 **12.** –80.00 **13.** 5.00 **14.** –2669.00 **15.** –84.00

Rankers Academy JEE

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	57	4	360	95	2	48	6535	1006	741	200
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	466	125	800	4	56	4	4	718	82	4
Que.	21	22	23	24	25					
Ans.	2	2	2	82	1					

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	
Ans.	A	D	D	C	AB	90.39	

For More Join: @IITJEE_Advanced